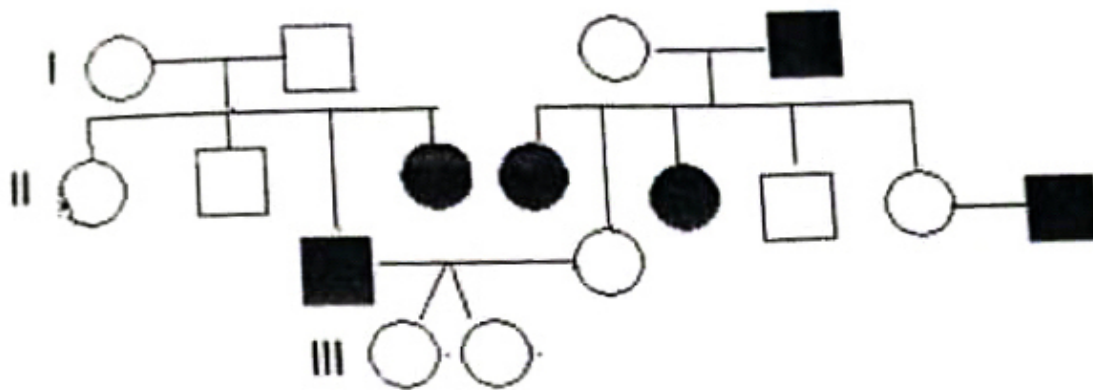


MEDICAL GENETICS

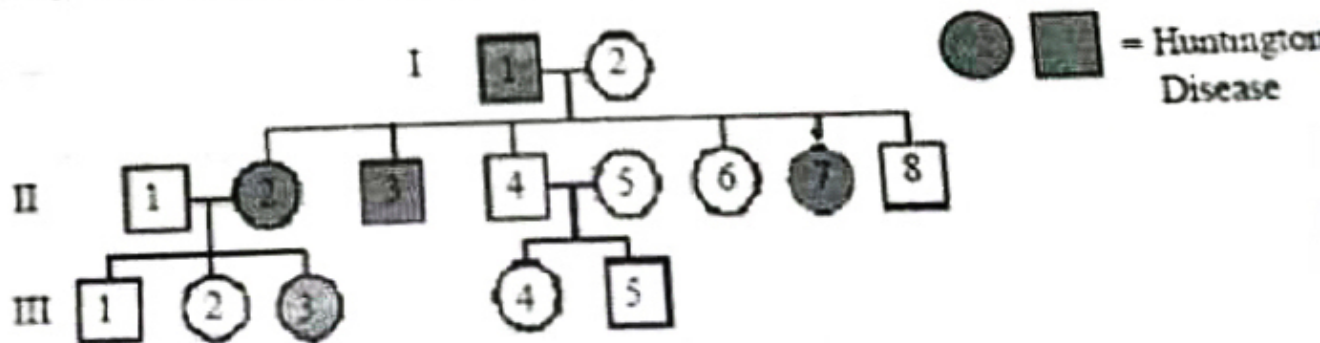
Read the questions carefully and Circle the right Answer

1. What pattern of inheritance is shown in the pedigree?



- ☒ A. Autosomal dominant
☐ B. Autosomal recessive
☐ C. Sex linked dominant
☐ D. Sex linked recessive

2. What pattern of inheritance is shown in the pedigree?



- ☒ A. Autosomal dominant
☐ B. Autosomal recessive
☐ C. Sex linked dominant
☐ D. Sex linked recessive

22. Umbilical vein carries
☒ A. Oxygenated blood from placenta to fetus
B. Deoxygenated blood from fetus to placenta
C. Deoxygenated blood from placenta to fetus
D. Oxygenated blood from fetus to placenta

Mention the types of Placenta and describe one of them;

23. Diffuse Placenta
24. Cotyledonary Placenta
25. Zonary Placenta
26. Discoid Placenta
27. Hemochorial Placenta
28. Epitheliochorial Placenta

Discoid Placenta is the chorionic villi are arranged in a disc-like shape. This type of placenta is found in primates and humans.

29. Which of the following hormone(s) is/are produced by the corpus luteum?

- A. progesterone only
☒ B. progesterone and estrogen only
C. progesterone, estrogen, and inhibin only
D. progesterone, estrogen, inhibin, and relaxin only

30. In the fetal circulation, the highest oxygen content occurs in which of the following?

- A. aorta
B. ductus arteriosus
☒ C. ductus venosus
D. superior vena cava

16. The small fissures between the lobes of the placenta are called

- A. raphes
- ☒ B. sulci
- C. gyri
- D. lacunae

17. The vertex is the region bounded by

- ☒ A. anterior and posterior fontanelle and parietal eminence
- B. parietal eminence, anterior fontanelle and occipital protuberance
- C. mentum, glabella and malar bones
- D. glabella, anterior fontanelle and temporal bones

18. The fertilized female ovum implants in the uterine lining this stage

- A. morula
- B. trophoblast
- C. secondary oocyte
- ☒ D. blastocyst

19. The function of the testis is to

- ☒ A. produce spermatozoa
- B. b.secrete testosterone
- C. produce fsh
- D. secrete lubricant for spermatozoa

20. Which of the following is not a function of the placenta?

- A. A nutrient transfer
- B. hormone production
- ☒ C. urine formation
- D. exchange of gases

21. Risk factors for placenta previa

- A. Multiparity
- B. Advanced maternal age
- C. Prior cesarean sections
- ☒ D. All of the above
- E. A and C

3. Lamarck hypothesized that species evolved

- A. From previous species
- B. Randomly
- C. By mating between different species
- ☒ D. Due to changes in the inherited DNA sequence

4. Why Lamarck is considered an influential figure in the field of evolution?

- ☒ A. He was the first to try to explain how new species evolve.
- B. He discovered DNA as the mechanism of inheritance.
- C. He cut the tails off mice to demonstrate acquired characteristics.
- D. He bred giraffes to have long necks.

5. A chromosome with a very short arm and a very long arm is referred to as

- A. Metacentric
- B. telocentric
- ☒ C. Acrocentric
- D. Sub-metacentric

6. A chromosome is the thickest during

- A. anaphase
- B. prophase
- C. interphase
- ☒ D. metaphase

7. If Bb is a gene pair of an individual then the alleles for this gene pair are

- A. A and B
- B. a and A
- ☒ C. a and b
- D. b and B

8. Each of the chromosome at the anaphase stage of a bone marrow cell in our body has

- A. one chromatid
- ☒ B. two chromatids
- C. several chromatids
- D. no chromatids

9. During cell division, spindle fibers attach to which part of chromosomes?

- ☒ A. Primary constriction
- B. Secondary constriction
- C. Chromomere

D. Chromatid

10. Apart from sex chromosome, which is the other type of chromosome?

- ☒ A. Autosome
- B. Cytosome
- C. Both A and B
- D. Protosome

11. Chromosomes contain information for inheritance of features from parents to next generation in the form of

- ☒ A. DNA
- B. RNA
- C. Proteins
- D. Chromatin

12. Chromosome carry which helps in the inheritance or transfer of characters from the parents to the offspring.

- ☒ A. Genes
- B. RNA
- C. Enzymes
- D. None of the above

13. The sex chromosomes of plants were first discovered in

- A. Algae
- B. Fungi
- ☒ C. Pteridophyta
- D. Flowering plants

14. In man the normal number of chromosomes

- A. 42
- B. 44
- C. 46
- D. 48

15. Any change in the nucleotide sequence of the DNA of a gene is called

- A. a mutation.
- B. an advantage.
- C. a codon.
- D. an anticodon.
- E. a translocation.